Exp no: 08

28.09.26

Nmap Live host Discovery on Tryhackme Platform.

Aim :-

To discover live hosts on a network using Nmap host-discovery scans - ARP scans, ICMP, and TCP/UDP ping scan - and to understand which protocol operates at which layer of the TCP/IP model, and when each one applies.

1) Introduction.

Nmap ("network Mapper") is a free, open-source network-scanning tool created by Gordon Lyon and first released in 1997. It is the industry standard tool.

* ARP (link layer)
* ICMP (network layer)
* TCP (Transport layer).
* UDP (Transport layer).

2) Learning objectives

* understand why host discovery must precede port scanning.
* learn how ARP scans works at the link layer using broadcast requests.

) Prerequisites:.

* A TryHackme account with access to the Nmap live host discovery" room

* A basic Linux terminal (Nmap comes re-installed on the Attackbox and on 'Kali' nux).

Background Concepts.

.1 Network segments and subnetworks.

A network segment is a group of computer nected by a shared physical medium - an hernet switch or a Wi-fi access point.

2 Target specification in Nmap.

* A list : nmap 10.10.1.5 scanme.nmap.org example.com.

* A range: nmap 10.11.12.15-20

* A subnet / CIDR: nmap 10.10.1.0/24.

) Lab setup or Tryhackme.

1. open the room
2. Read Task 1.
3. In Task 4.
4. click Start Attack Box.
5. Note the connection - IP.

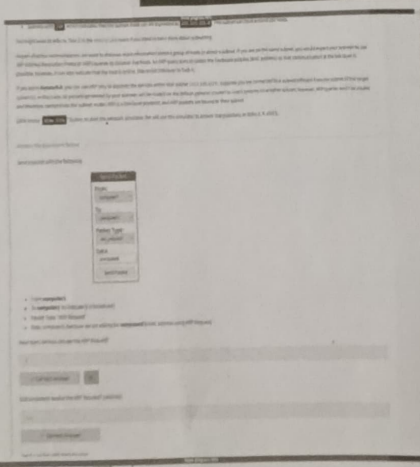
```
pentester@TryHackMe$ sudo nmap -PU -sn 10.10.68.220/24
Starting Nmap 7.92 ( https://nmap.org ) at 2021-09-02 13:45 EEST
Nmap scan report for 10.10.68.52
Host is up (0.10s latency).
Nmap scan report for 10.10.68.121
Host is up (0.10s latency).
Nmap scan report for 10.10.68.125
Host is up (0.14s latency).
Nmap scan report for 10.10.68.134
Host is up (0.096s latency).
Nmap scan report for 10.10.68.220
Host is up (0.11s latency).
Nmap done: 256 IP addresses (5 hosts up) scanned in 9.20 seconds
```

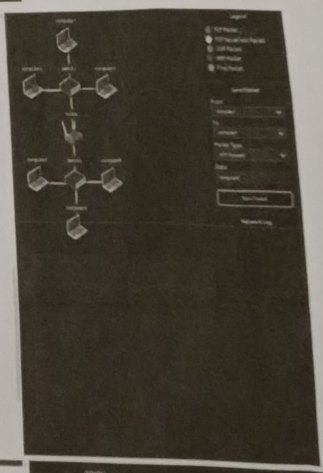
```
pentester@TryHackMe$ sudo nmap -PE -sn 10.10.210.6/24

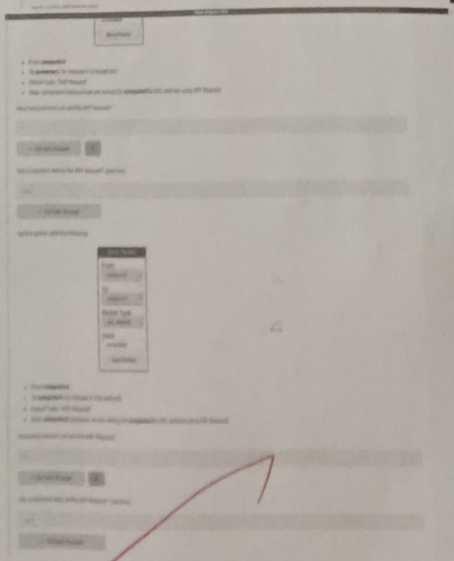
Starting Nmap 7.60 ( https://nmap.org ) at 2021-09-02 07:12 BST
Nmap scan report for ip-10-10-210-75.eu-west-1.compute.internal (10.10.210.75)
Host is up (0.00013s latency).
MAC Address: 02:83:75:3A:F2:89 (Unknown)
Nmap scan report for ip-10-10-210-100.eu-west-1.compute.internal (10.10.210.100)
Host is up (-0.10s latency).
MAC Address: 02:63:D0:18:2D:CD (Unknown)
Nmap scan report for ip-10-10-210-165.eu-west-1.compute.internal (10.10.210.165)
Host is up (0.00025s latency).
MAC Address: 02:59:79:4F:17:B7 (Unknown)
Nmap scan report for ip-10-10-210-6.eu-west-1.compute.internal (10.10.210.6)
Host is up.
Nmap done: 256 IP addresses (4 hosts up) scanned in 3.12 seconds
```

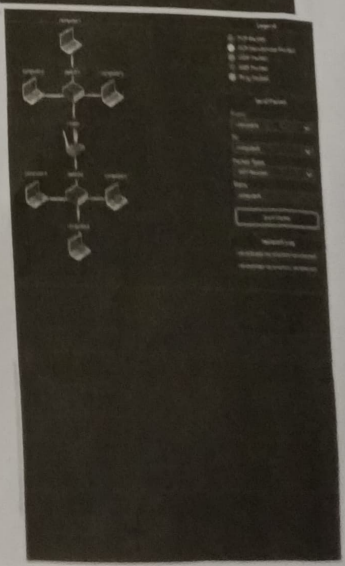
```
pentester@TryHackMe$ sudo nmap -PE -sn 10.10.68.220/24

Starting Nmap 7.92 ( https://nmap.org ) at 2021-09-02 12:16 EEST
Nmap scan report for 10.10.68.50
Host is up (0.12s latency).
Nmap scan report for 10.10.68.52
Host is up (0.12s latency).
Nmap scan report for 10.10.68.77
Host is up (0.11s latency).
Nmap scan report for 10.10.68.110
Host is up (0.11s latency).
Nmap scan report for 10.10.68.140
Host is up (0.11s latency).
Nmap scan report for 10.10.68.142
Host is up (0.11s latency).
Nmap scan report for 10.10.68.220
Host is up (0.11s latency).
Nmap scan report for 10.10.68.222
Host is up (0.11s latency).
Nmap done: 256 IP addresses (8 hosts up) scanned in 8.26 seconds
```









10) Masscan - A faster Alternative.

Masscan performs a similar discovery / port sweep but transmits packets far more aggressively, trading some accuracy ranges,

11) Reverse - DNS lookup:-

Reverse DNS (rDNS) resolves an IP address back to a hostname. Nmap performs rDNS lookups on live hosts by default.

Result:-

Live hosts in the given network were successfully identified using different nmap.